

On the meaning of “collision rate constants” for ion-molecule reactions: Association of hydrogen atoms with $C_6H_5^+$ and small alkyl radicals with $C_7H_7^+$ ions

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ABSTRACT

Limiting high pressure rate constants for the association of H with $C_6H_5^+$ and of CH_3 , C_2H_5 , and $n-C_3H_7$ radicals with $C_7H_7^+$ molecular ions are analyzed in terms of “rigidity factors” arising from the anisotropy of the interaction potentials and “collision rate constants” in the absence of anisotropy (the latter corresponding to phase space theory, PST). Model calculations based on ab initio potential energy surfaces show that the PST rate constants $k^{PST}(T)$ exceed collision rate constants from conventional ion-molecule capture theory (in this case given by Langevin rate constants k_L). They can be represented by $k^{PST}(T) = k_L + k_{h.sph.}$ where $k_{h.sph.}$ denotes hard-sphere collision numbers with collision radii r_0 . The r_0 derived from the modelled $k^{PST}(T)$ are related to properties of the interaction potentials. Applications to other ion-molecule reactions are proposed.

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1. Introduction

It is common practice to compare experimental rate constants k of bimolecular ion-molecule reactions with “collision rate constants”. In the case of polar reactants, the latter are conveniently estimated by the Su-Chesnavich rate constant $k_{S.Ch.}$, see ref. [1],

$$k_{S.Ch.}/k_L = 1 + 0.0967x + 0.0950x^2 \text{ for } x \leq 2 \quad (1)$$

$$= 0.4764x + 0.6200 \text{ for } x \geq 2 \quad (2)$$

where the Langevin rate constant k_L is given by

$$k_L = 2\pi q(\alpha/\mu)^{1/2} \quad (3)$$

α denotes the polarizability of the neutral reactant, q the charge of the ion, μ the reduced mass of the collision pair, and the parameter x is given by $x = \mu_D/(2\alpha k_B T)^{1/2}$ with the dipole moment μ_D of the neutral reactant. Eqs. (1) – (3) are the result of classical trajectory (CT) calculations for collision processes between two point masses

in a long-range charge-induced + permanent neutral dipole potential. Rotational quantum effects at low temperatures result in a decrease of k to values smaller than $k_{S.Ch.}$. Such effects are accounted for, e.g. by the statistical adiabatic channel model (SACM) [2–6], the average dipole orientation (ADO) [7,8] or the adiabatic capture centrifugal sudden approximation (ACCSA) [9,10], all essentially being equivalent [11,12]. Translational quantum effects become apparent at even lower temperatures, see e.g. ref. [13].

The comparison of eqs. (1) – (3) with experimental results showed a wide range of possibilities. Agreement with experiments, e.g., has been observed for reactions like $H_3^+ + HCN$ (see the analysis of this reaction in refs. [2,4]). Limiting high-pressure rate constants for ion-molecule associations, which correspond to collisional control, in many cases are smaller than given by eqs. (1) – (3), see e.g. the modelling in ref. [14]. On the other hand, rate constants k can also exceed $k_{S.Ch.}$, particularly in reactions with ionic clusters, see refs. [15–18], or in the attachment of electrons to neutral clusters [19–24].

Rate constants, k , larger than the collision rate constants, $k_{S.Ch.}$, have been interpreted in different ways. “Finite-target size effects”, e.g., have been considered for reactions of ionic clusters or for the attachment of electrons to neutral clusters [15–24]. Apart from contributions of quadrupole moments in addition to dipole

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moments of the neutral [5], influences of polarizabilities, dipole and quadrupole moments of the ions were also suggested for an explanation, see, e.g., ref. [15]; this accounts for changes of the potential between long-range (LR) and short-range (SR) electrostatic contributions. Alternatively, one may replace the SR electrostatic potential by a SR potential of valence character as discussed in the present article.

There are multiple influences of the potential on the collision rate constant k (if one understands by “collision” the occurrence of “capture”, i.e. the formation of an adduct able to react). There is first the influence of the minimum-energy path (MEP) potential $V(r)$. In addition, the anisotropy of the potential is important. While a LR polarization potential often is nearly isotropic, the LR potential may also be anisotropic, e.g., in the case of a LR ion + permanent dipole interaction. The latter is accounted for in $k_{S,Ch}$. [2]. The total (LR plus SR) anisotropy of the potential gives rise to “rigidity factors”, f_{rigid} , between zero and unity [14,25]. Anisotropy contributions to k are neglected in phase space theory (PST, see, e.g., ref. [26]). The aim of the present article is to investigate finite-target size effects in PST. As such effects are system-specific, we consider a series of reaction systems in order to obtain some generalizable results. We consider the example of high-pressure rate constants for association of small alkyl radicals to $C_7H_7^+$ cations and their representation by PST, i.e. putting rigidity factors equal to unity. Likewise, we consider the association of hydrogen atoms to $C_6H_5^+$ cations. We rely on SACM/CT calculations from refs. [6,14,27], employing SR valence/LR electrostatic switching potentials, in part based on ab initio calculations of the potential energy surfaces (PESs) such as described in refs. [28,29] and extended in the present work. The advantage of building on the results of refs. [6,14,27] lies in the fact that rigidity factors in this work have been calculated separately and rate constants $k^{PST}(T)$ for PST have been determined. In this way, the contribution of finite-target size effects to $k^{PST}(T)$ can directly be explored.

2. PST rate constants for association reactions in their high-pressure limits

We started with the $n-C_3H_7 + C_7H_7^+ \rightarrow C_{10}H_{14}^+$ ion-molecule association system. We extended earlier ab initio calculations of the PES from refs. [27,29]. In particular, the properties of the distance dependence and the anisotropy of the SR valence/LR polarization switching potential used in ref. [27] were controlled. Our calculations were made on the $\omega B97X-D/6-311++G(3df,3pd)//\omega B97X-D/6-311++G(d,p)$ DFT level [30,31] of theory and compared with composite ab initio models such as G4MP2 as well as the less accurate B3LYP/6-31G(d) calculations (the latter also used in ref. [29]). Fig. 1 illustrates the r -dependence of the MEP electronic potential $V(r)$. While the various calculations agree fairly well in the $V(r)/D_e$ representations at small values of r , i.e. in the SR valence range, larger differences are observed at large r , when the LR polarization potential takes over (one should note that in the present work r denotes the C–C distance; it differs from the center-of-mass distance r_{com} , which is of importance when the simplification of point masses used for eqs. (1)–(3) is abandoned, see below). Larger differences are also obtained for the electronic bond dissociation energy D_e (or the corresponding reaction enthalpy ΔH_0° , the latter obtained as 195.4 with the $\omega B97X$ functional, 193.3 with the G4MP2, and 161.6 kJ mol^{−1} with the B3LYP calculations).

Fig. 1 includes a fit of the numerical results for the SR valence potential by a Morse expression

$$V^{SR}(r) \approx D_e [1 - \exp(-\beta_e(r - r_e))]^2 \quad (4)$$

with the parameters $r_e = 1.572$ Å and $\beta_e = 1.60$ Å^{−1}. Following ref.

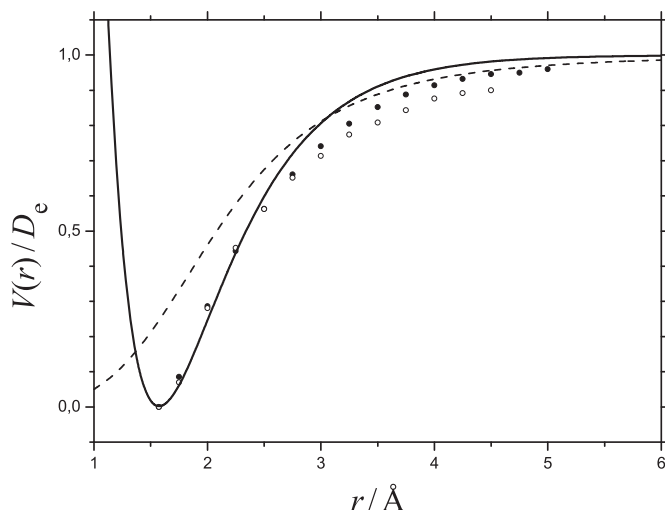


Fig. 1. MEP electronic potential $V(r)$ for the reaction $n-C_3H_7 + C_7H_7^+ \rightarrow C_{10}H_{14}^+$ (r : C–C distance, open circles: B3LYP/6-31G(d) calculations, filled circles: $\omega B97X-D/6-311++G(3df,3pd)//\omega B97X-D/6-311++G(d,p)$ calculations, $D_e = 203.3$ kJ mol^{−1} for the filled circles, 156.9 kJ mol^{−1} for the open circles; dashed line: LR potential of eq. (6), solid line: SR potential of eq. (4) with $D_e = 203.3$ kJ mol^{−1}, $r_e = 1.572$ Å, and $\beta_e = 1.60$ Å^{−1}).

[6], the LR polarization potential

$$V^{LR}(r) \approx D_e - \alpha q^2 / 2 r_{com}^4 \quad (5)$$

is modified into the form

$$V^{LR}(r) \approx D_e - \alpha q^2 / [2(r^4 + \alpha q^2 / 2D_e)] \quad (6)$$

(where the center-of-mass distance r_{com} between the two mass points is replaced by the length of the C–C bond r , and a correction term $\alpha q^2 / 2D_e$ was suggested for the denominator). Fig. 1 includes $V^{LR}(r)$ (with $\alpha = 5.8$ Å³). One notes that the results for the $\omega B97X-D$ functional satisfactorily approach the LR potential of eq. (6). A suitable switching between $V^{SR}(r)$ and $V^{LR}(r)$ has been described in ref. [6] and needs not to be repeated here. The resulting SR/LR switching potential then serves as the basis for the calculation of the rate constants which, after the present slight improvement of the potential, gave essentially unchanged results.

The anisotropy of the PES is characterized by the transition mode frequencies along the MEP of the system. Fig. 2 shows the results from the B3LYP/6-31G(d) and $\omega B97X-D/6-311++G(d,p)$ calculations. As usual, an exponential decrease of $\nu(r)$ is observed. On average, this decrease is described by $\nu_i(r) = \nu_i(r_e) \exp(-\alpha_e(r - r_e))$ with $\alpha_e = 0.66$ Å^{−1}. This result is essential for the determination of the PST rate constants in the absence of anisotropy contributions. The thermal rigidity factors, $f_{rigid}(T)$, from the detailed SACM/CT calculations of ref. 27 and from the approximate calculations following refs. [32,33] were found to agree satisfactorily well. We, therefore, can proceed with the analysis of the modelled rate constants $k^{PST}(T) = k(T)/f_{rigid}(T)$ (one should note that the modelled $k(T)$ were consistent with experimental data, see refs. [6, 14, 27]).

The reduction of the experimental k to values below $k^{PST}(T)$ by the rigidity factors is quite substantial. This is illustrated in Fig. 3 where the temperature dependences of k and $k^{PST}(T)$ are shown between 10 and 1000 K. $k(T)$ starts from a low-temperature limit, approximately equal to $k^{PST}(T)$ and given by the Langevin rate constant, $k_L = 1.08 \times 10^{-9}$ cm³ s^{−1}. It then increasingly falls below $k^{PST}(T)$ as the temperature increases. On the other hand, the modelled $k^{PST}(T)$ slightly increases with increasing temperature.

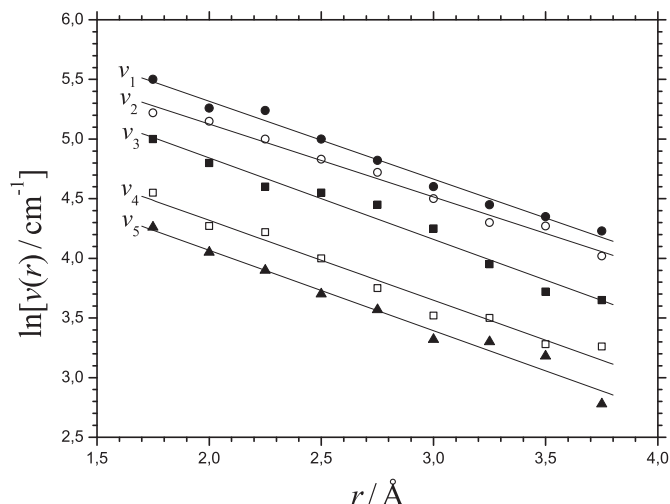


Fig. 2. Transitional mode frequencies for the reaction $n\text{-C}_3\text{H}_7 + \text{C}_7\text{H}_7^+ \rightarrow \text{C}_{10}\text{H}_{14}^+$ along the MEP (from $\omega\text{B97X-D/6-311++G(d,p)}$ calculations, see Fig. 1; lines: representation by $\nu_i(r) = \nu_i(r_e) \exp(-\alpha_{ei}(r - r_e))$ with fitted values $\alpha_{ei} = 0.66, 0.61, 0.69, 0.67, 0.69 \text{ Å}^{-1}$ from top to bottom).

This increase is systematic. It is documented in Table 1 where the association reactions of C_7H_7^+ with CH_3 , C_2H_5 , and $n\text{-C}_3\text{H}_7$ and of H with C_6H_5^+ are compared. The table represents the modelled $k^{\text{PST}}(T)$ in the form

$$k^{\text{PST}}(T) \approx k_L + k_{h,\text{sph}} \quad (7)$$

with hard-sphere collision numbers $k_{h,\text{sph}}$ given by

$$k_{h,\text{sph}} = \pi r_{\text{com},0}^2 (8 k_B T / \pi \mu)^{1/2} \quad (8)$$

$k^{\text{PST}}(T)$ in the table is compared with k_L . The representation with eqs. (7) and (8) follows the suggestions of refs. [23,24], made for reactions with finite-target size reactants. Fitting the modelled $k^{\text{PST}}(T)$ to eqs. (7) and (8) leads to values of $r_{\text{com},0}$ of about 4.2, 5.0, and 5.1 Å for the association reactions forming $\text{C}_8\text{H}_{10}^+$, $\text{C}_9\text{H}_{12}^+$, and $\text{C}_{10}\text{H}_{14}^+$, respectively, while $r_{\text{com},0} \approx 3.0 \text{ Å}$ is obtained for $\text{H} + \text{C}_6\text{H}_5^+$. In the next section, we relate these values of “effective collision radii” $r_{\text{com},0}$ to the potentials $V(r)$ underlying the SACM/CT calculations of k and $k^{\text{PST}}(T)$.

2.1. Interpretation of “effective collision radii” $r_{\text{com},0}$

The transitional mode frequencies $\nu(r)$ shown in Fig. 2, in combination with the electronic potential $V(r)$ of Fig. 1, led to rigidity factors $f_{\text{rigid}}(T)$ and $f_{\text{rigid}}(EJ)$ such as determined in the framework of SACM/CT in refs. [6,14,27] and illustrated in Fig. 3 for the reaction $n\text{-C}_3\text{H}_7 + \text{C}_7\text{H}_7^+ \rightarrow \text{C}_{10}\text{H}_{14}^+$. The rigidity factors are not further considered here. Instead, we investigate how, after putting $f_{\text{rigid}}(T)$ and $f_{\text{rigid}}(EJ)$ equal to unity, $k^{\text{PST}}(T)$ changes when the simple LR polarization potential is replaced by the SR/LR switching potential and when mass points are replaced by finite-target size reactants.

Most importantly, the replacement of the simple LR polarization

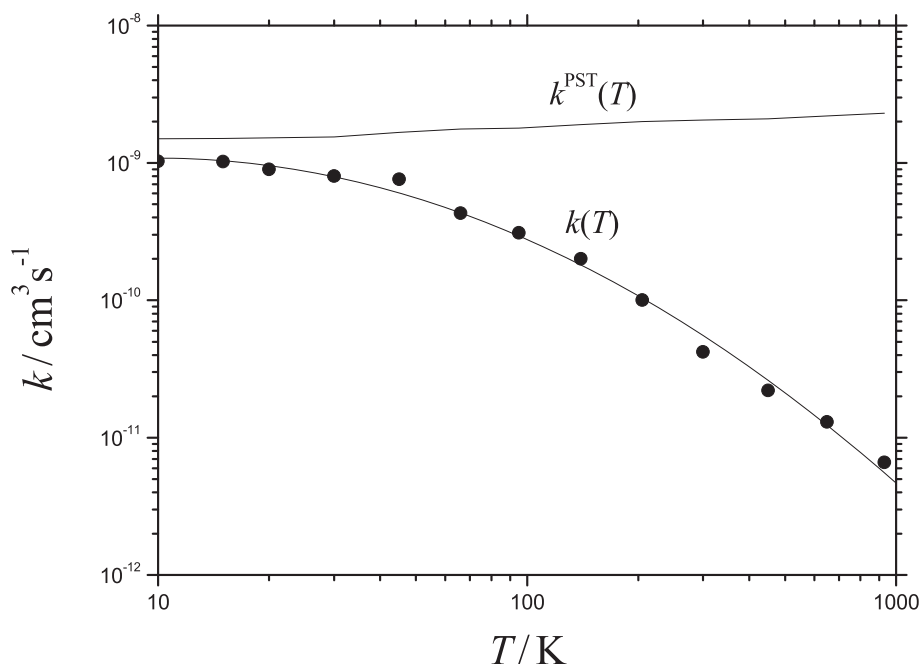


Fig. 3. Complete rate constants $k(T)$ and PST rate constants $k^{\text{PST}}(T)$ (rigidity factors $f_{\text{rigid}}(T) = 1$) for the reaction $n\text{-C}_3\text{H}_7 + \text{C}_7\text{H}_7^+ \rightarrow \text{C}_{10}\text{H}_{14}^+$ (results from ref. [27], see text).

Table 1

Rate constants for ion-molecule association reactions (k_L = Langevin rate constants, k^{PST} = rate constants from phase space theory, PST; $r_{\text{com},0}$ = hard-sphere collision radii from eqs. (7) and (8), $r_{\text{com},e}$ = equilibrium lengths of the forming bonds in eq. (4), $r_{\text{com},0}$ and $r_{\text{com},e}$ are center-of-mass coordinates).

Reaction	$k_L/10^{-9} \text{ cm}^3 \text{ s}^{-1}$	$k^{\text{PST}}(1000 \text{ K})/10^{-9} \text{ cm}^3 \text{ s}^{-1}$	$r_{\text{com},0}/\text{Å}$	$r_{\text{com},e}/\text{Å}$
$\text{H} + \text{C}_6\text{H}_5^+$	1.92	4.2	3.0	2.5
$\text{CH}_3 + \text{C}_7\text{H}_7^+$	1.05	2.3	4.2	3.5
$\text{C}_2\text{H}_5 + \text{C}_7\text{H}_7^+$	1.04	2.4	5.0	4.7
$n\text{-C}_3\text{H}_7 + \text{C}_7\text{H}_7^+$	1.09	2.3	5.1	5.2

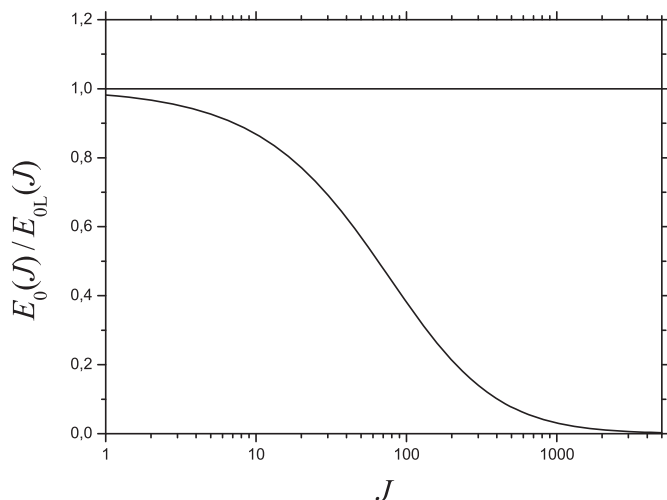


Fig. 4. Centrifugal barriers $E_0(J)$ for the reaction $n\text{-C}_3\text{H}_7 + \text{C}_7\text{H}_7^+ \rightarrow \text{C}_{10}\text{H}_{14}^+$ (lower line: results for the MEP potential $V(r)$ of Fig. 1, upper line and $E_{0L}(J)$: results for the Langevin model, see ref. [2]).

potential by the SR/LR switching potential modifies the centrifugal barriers $E_0(J)$ to be overcome in the collision process. A decrease of $E_0(J)$ (at a given J) results in an increase of the rate constants $k^{\text{PST}}(T)$. This change of the $E_0(J)$ is due to two effects. (i) When the SR valence potential at small r is more attractive than the LR polarization potential, the two potentials cross at some distance r_{sw} ; while the centrifugal barriers for small J are located at $r > r_{\text{sw}}$, with increasing J they move into the range $r < r_{\text{sw}}$. The change of $V(r)$ from the LR potential to the MEP SR/LR switching potential then results in a lowering of the centrifugal barriers. (ii) Replacing mass-point by finite-target size reactants modifies the effective moments of inertia of the collision pair; this effect can be illustrated, e.g., by inspecting the rotational constants $0.5(B + C)$ of the collision pair

along the MEP. We illustrate the two effects in Figs. 4 and 5.

First, Fig. 4 compares $E_0(J)$ for the $\text{C}_{10}\text{H}_{14}^+$ system (from the SACM/CT calculations of ref. 27) with the corresponding results underlying k_L (see ref. 2). $E_0(J)$ in the SACM/CT calculations was derived to be $E_0(J)/hc \approx 2.47 \times 10^{-7} J^4 / (1 + 0.0181 J^{0.879} + 3.67 \times 10^{-4} J^{1.601}) \text{ cm}^{-1}$. On the other hand, $E_{0L}(J)/hc \approx 2.47 \times 10^{-7} J^4 \text{ cm}^{-1}$ corresponds to the Langevin model [2]. Fig. 4 shows that $E_0(J)$ increasingly falls below $E_{0L}(J)$ as J is increased. One might try to express the ratio k/k^{PST} in terms of the corresponding centrifugal partition functions

$$Q_{\text{cent}}^* \approx \sum [2J \exp(-E_0(J) / k_B T)] \quad (9)$$

However, this would not be sufficient; $k^{\text{PST}}(T)$ also includes [26] a ratio of rotational partition functions of the “activated complex” and the reactants, Q_{rot}^* and Q_{rot} , respectively. This ratio is equal to unity in the Langevin model while it differs from unity in the present case. $Q_{\text{rot}}^*(r)$ here is based on the rotational constant $0.5(B + C)$ as a function of r along the MEP which, in the present work and for the $\text{C}_{10}\text{H}_{14}^+$ system, was calculated to be $0.5(B + C) \approx 0.0194 \text{ cm}^{-1} / [1 + 0.122(r - 1.572 \text{ Å}) + 0.0546(r - 1.572 \text{ Å})^2]$ which, at large r , approaches an r^{-2} dependence while the short-range deviations from this dependence are caused by the valence part of the potential. Fig. 5 illustrates the r -dependence of $0.5(B + C)$, which considerably differs from that of a point-mass model (by combination with $V(r)$ this led to the $E_0(J)$ given above). The access to k/k^{PST} via Q_{cent}^* , $Q_{\text{rot}}^*(r)$, and Q_{rot} finally requires a variational approach (which essentially is equivalent to the SACM/CT calculations of Fig. 3). While the SACM/CT and variational TST calculations lead to quantitative representations of the “effective collision radii” $r_{\text{com},0}$ of eqs. (4) and (5), a more qualitative interpretation may also provide insight. The empirical values for $r_{\text{com},0}$ of Table 1 apparently are all close to the equilibrium bond lengths $r_{\text{com},e}$ of the adduct potentials also given in the table. When one now replaces the SR/LR switching potential by an expression

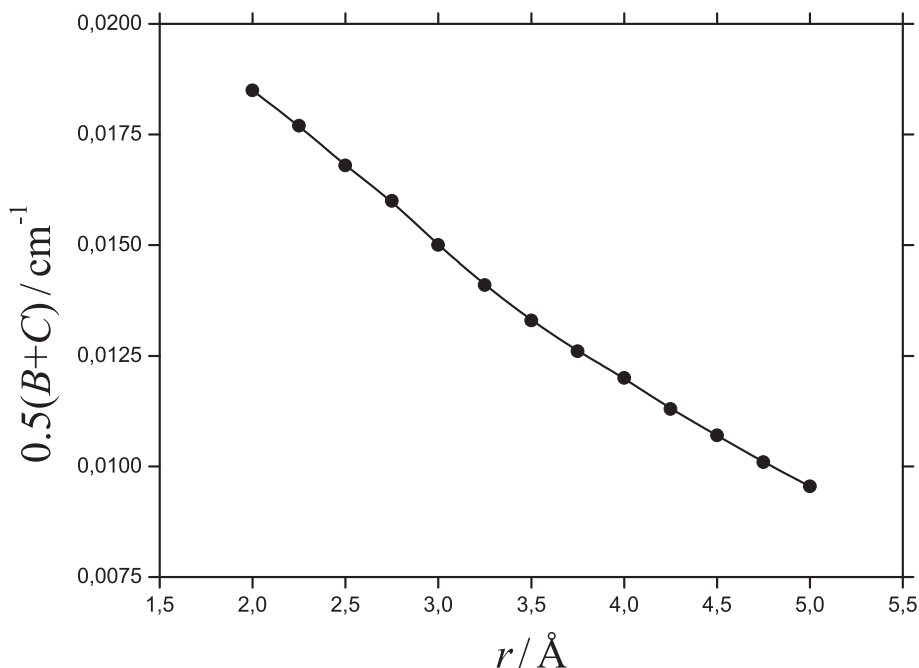


Fig. 5. Rotational constants along the MEP of the reaction $n\text{-C}_3\text{H}_7 + \text{C}_7\text{H}_7^+ \rightarrow \text{C}_{10}\text{H}_{14}^+$ (filled points from $\omega\text{B97X-D/6-311++G(d)}$ calculations, see Fig. 1; line: expression given in the text).

$$V(r_{\text{com}})/D_e \approx 1 - (\alpha q^2/2 D_e) / [r_{\text{com}}^2 (r_{\text{com}}^2 - r_{\text{com},0}^2)] \quad (10)$$

the corresponding collisional rate constant $k^{\text{PST}}(T)$ according to refs. [23,24] takes the form of eqs. (7) and (8). As the “hard-sphere values” $r_{\text{com},0}$ are not far from the equilibrium bond lengths $r_{\text{com},e}$ of the SR valence potentials, the appropriateness of eqs. (7) and (8) appears plausible, although $V(r_{\text{com}})$ from eq. (10) substantially differs from $V(r)$ used before. The finer details of the SR/LR switching potential apparently are more important for the calculation of the rigidity factors than for the determination of $k^{\text{PST}}(T)$. The four reactions included in Table 1 document the similarity of the modelled $r_{\text{com},0}$ and the bond lengths $r_{\text{com},e}$ of the SR valence potentials.

3. Conclusions

The present work has analyzed limiting high pressure rate constants for association processes of $\text{C}_6\text{H}_5^\pm$ and $\text{C}_7\text{H}_7^\pm$ ions with small radicals. After removing the “rigidity factors” $f_{\text{rigid}}(T)$, which reduce the rate constants as a consequence of the anisotropy of the potential, the remaining PST rate constants $k^{\text{PST}}(T)$ all exceed collision rate constants from conventional ion-molecule capture theory. The added contributions correspond to hard-sphere collision numbers, being proportional to $T^{1/2}$. The effective hard-sphere collision radii $r_{\text{com},0}$, to a good approximation, can be identified with the equilibrium lengths $r_{\text{com},e}$ of the forming bonds such as characterized by the SR valence potentials.

Rate constants of ion-molecule reactions, which exceed “collision rate constants” given by the Su-Chesnavich equations (1)–(3) (or the Langevin rate constant of eq. (3) in the absence of a permanent dipole moment), thus can be rationalized by finite-reactant size and SR-valence potential effects, as long as the anisotropy of the SR potential is not too large. Neglecting the anisotropy effects, the “collision rate constants” are given by the sum of Su-Chesnavich type rate constants and hard-sphere collision numbers related to the SR valence potentials. It should be mentioned that similar “finite-target size effects” are also relevant for electron attachment, even to molecules as small as SF_6 (see ref. [34]).

The conclusions from the present study should be transferable to other ion-molecule reactions. Substantial anisotropy in the valence forces reduce the rate constant to values below $k^{\text{PST}}(T)$. If, on the other hand, the SR potentials are not very attractive, $r_{\text{com},0}$ should correspond to the sum of the hard sphere radii of the neutral and ionic reactants, and eqs. (7) and (8) should apply. If the polarizability (or dipole moment) of the neutral is only small, the hard sphere collision number may dominate $k^{\text{PST}}(T)$, such that $k \approx k_{\text{h.sph.}} \propto T^{1/2}$. Measurements of the temperature dependence of k thus provide insight into the relevant SR/LR potential of the interaction. This general conclusion certainly will also require studies of more systems than analyzed in the present work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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